Customer Shopping Analysis from Data to Dashboard

An End-to-End Data Analytics Project

1. Project Overview

This project analyses customer shopping behaviour using transactional retail data. The dataset contains **3,900 purchase records** across multiple product categories. The objective is to examine spending patterns, customer segmentation, product performance, discount usage, and subscription behaviour using a structured data analytics workflow.

The analysis was carried out using **Python for data exploration and preparation**, **PostgreSQL for SQL-based analysis**, and **Power BI for dashboard visualization**.

2. Dataset Summary

- **Rows**: 3,900
- **Columns**: 18

Data Attributes:

- **Customer demographics:** Age, Gender, Subscription Status
- **Purchase details:** Item Purchased, Category, Purchase Amount, Season, Size, Color
- **Shopping Behaviour:** Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type
- **Missing Data:** 37 missing values in the Review Rating column.

3. Data Exploration and Preparation (Python)

Initial data preparation and exploratory analysis were conducted in Python using pandas which include the following steps:

- **Data Loading:** Imported the dataset using pandas.
- **Initial Exploration:** Used df.info() to check structure and df.describe() for summary statistics.
- **Missing Data Handling:** Checked for null values and imputed missing values in the Review Rating column using the median rating of each product category.
- **Column Standardization:** Renamed columns to **snake case** for better consistency.

- **Feature Engineering:**
  - Created **age_group** column by binning customer ages.
  - Created **purchase_frequency_days** column from purchase data.
- **Data Consistency Check:** Verified if **discount_applied** and **promo_code_used** were redundant; dropped **promo_code_used**.
- **Database Integration:** Exported cleaned dataset from Python and loaded into a PostgreSQL database **(Dockerised)** to enable structured SQL-based analysis and aggregation.

	customer_id	age	gender	item_purchased	category	purchase_amount	location	size	color	season	review_rating
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7

review_rating	subscription_status	shipping_type	discount_applied	previous_purchases	payment_method	frequency_of_purchases	age_group	purchase_frequency_in_days
3.1	Yes	Express	Yes	14	Venmo	Fortnightly	50+ Senior	15
3.1	Yes	Express	Yes	2	Cash	Fortnightly	0-25 Young Adult	15
3.1	Yes	Free Shipping	Yes	23	Credit Card	Weekly	36-50 Middle-aged	7
3.5	Yes	Next Day Air	Yes	49	PayPal	Weekly	0-25 Young Adult	7
2.7	Yes	Free Shipping	Yes	31	PayPal	Annually	36-50 Middle-aged	365

Figure 1: Preview of Cleaned Dataset After Data Preparation

4. Analysis using SQL (PostgreSQL)

The analysis focused on descriptive business metrics which included the following:

1. **Revenue by Gender:** Comparison of total revenue generated by male and female customers

gender	sales_amount
Female	75191
Male	157890

2. **High-Spending Discount Users:** Identification of customers who used discounts but spent above the average purchase amount

customer_id	purchase_amount
2	64
3	73
4	90
7	85
9	97
12	68
13	72
16	81
20	90
22	62

10 / 839 rows (0.002 s) Edit, Explain

3. **Top 5 Products by Rating:** Products with the highest average **review_rating**

item_purchased	average_product_rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

4. **Shipping Type Comparison:** Average purchase amounts for Standard vs Express shipping

shipping_type	average_purchase_amount
Standard	58.46
Express	60.48

5. **Subscribers vs Non-Subscribers:** Comparison of average spend and total revenue

subscription_status	total_customers	average_spending	total_sales_amount
No	2847	59.87	170436
Yes	1053	59.49	62645

6. **Discount-Dependent Products:** Products with the highest percentage of discounted purchases

item_purchased	percentage_of_purchases_with_discount
Hat	50.00
Sneakers	49.00
Coat	49.00
Sweater	48.00
Pants	47.00

7. **Customer Segmentation:** Classification into New, Returning, and Loyal customers using **previous_purchases**

customer_segment	number_of_customers
Loyal	3116
New	83
Returning	701

8. **Repeat Buyers and Subscriptions:** Subscription status among customers with more than five previous purchases

subscription_status	repeat_orders
No	2518
Yes	958

9. **Top 3 Products per Category:** Most purchased items within each category

item_rank	category	item_purchased	total_orders
1	Accessories	Jewelry	171
2	Accessories	Sunglasses	161
3	Accessories	Belt	161
1	Clothing	Blouse	171
2	Clothing	Pants	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145
1	Outerwear	Jacket	163

10. **Revenue by Age Group:** Total revenue contribution by each age group

age_group	total_revenue
50+ Senior	88480
36-50 Middle-aged	65629
26-35 Adult	44342
0-25 Young Adult	34630

5. Power BI Dashboard Development

An interactive dashboard was developed in **Power BI** to visually present the results of the SQL analysis and support exploratory reporting.

The dashboard includes:

- **KPI cards** displaying the total number of customers (3.9K), average purchase amount ($59.76), and average review rating (3.75)
- **Revenue by category** visual highlighting contribution from Clothing, Accessories, Footwear, and Outerwear
- **Subscription status distribution** showing the proportion of subscribed and non-subscribed customers
- **Interactive slicers** for gender and age group to enable demographic-based filtering
- A clean, single-page layout designed for high-level overview and quick comparison

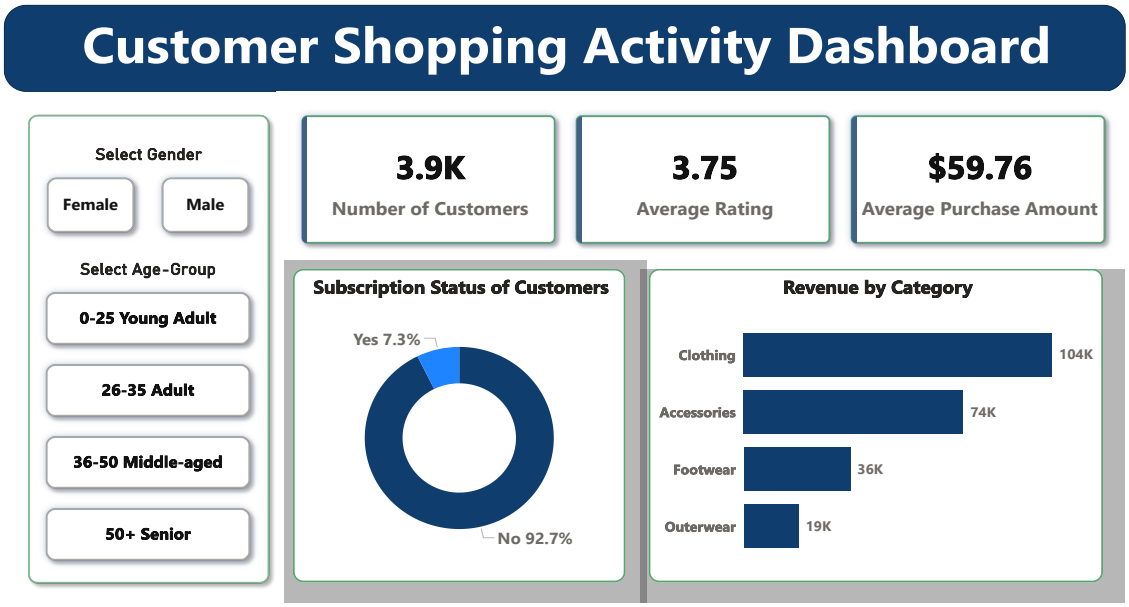

Figure 2: Power BI Dashboard for Customer Shopping Data

6. Tools and Technologies Used

- **Python (pandas):** Data cleaning and feature engineering
- **PostgreSQL:** Analytical SQL queries
- **Docker:** Database deployment
- **Power BI:** Dashboard and visual reporting
- **GitHub:** Version control

7. Business Recommendations

The following recommendations are derived from descriptive analysis and dashboard observations:

- **Boost Subscriptions:** Promote subscription benefits, particularly to repeat buyers
- **Customer Loyalty Program:** Encourage returning customers to transition into the loyal segment
- **Review Discount Policy:** Monitor products with high discount dependency to balance sales volume and margins
- **Product Positioning:** Highlight high-rated and high-volume products in promotions
- **Targeted Marketing:** Focus campaigns on high-revenue age groups and customers using express shipping